

VIRTUAL SYSTEMS AND SERVICES

Weekly Lecture Schedule • Fall 2026

Bachelor of Science in Information Technology • *From Hypervisors to the Cloud-Native Stack*

Modernised from the HEC 2017 outline: the same hypervisor foundations, carried forward into containers, Kubernetes and cloud virtualization

CREDIT HOURS	DURATION	PREREQUISITE	CONTENTS
3 (2–1): 2 lecture h + 1 lab h a week	16 teaching weeks; midterm Week 8, project due Week 16	Programming Fundamentals (HEC 2017 outline)	23 chapters in 6 parts; hypervisors, containers and cloud VMs

📅	📖 Lecture topics	🔧 Laboratory (1 h)	📋 Assessment
1	1 What Virtualization Is – abstraction of compute, storage and network; why virtualize; from IBM VM/370 to the cloud 2 Virtualization Architectures – Type-1 vs Type-2 hypervisors, full virtualization, paravirtualization, hardware-assisted virtualization	Install a Type-2 hypervisor; a first VM and snapshot	—
2	3 The Virtual Machine Monitor – the Popek–Goldberg theorem, trap-and-emulate, binary translation, ring deprivileging	Trace a privileged instruction; trap-and-emulate by hand	—
3	4 Hardware Support for Virtualization – Intel VT-x/VT-d, AMD-V/AMD-Vi, extended/nested page tables, the IOMMU	Check VT-x/AMD-V flags; enable nested virtualization	—
4	5 Major Hypervisor Platforms – VMware ESXi/vSphere, Microsoft Hyper-V, KVM/QEMU, Xen; choosing between them	The same workload on ESXi, Hyper-V and KVM	—
5	6 CPU and Memory Virtualization – vCPU scheduling, overcommit, ballooning, transparent page sharing, NUMA	Ballooning and overcommit under memory pressure	—
6	7 Paravirtualization and Device Virtualization – virtio and PV drivers, emulated vs passthrough devices, SR-IOV, VFIO	virtio-net against an emulated NIC, throughput compared	Quiz 1 (Ch 1–7)
7	8 From Virtual Machines to Containers – namespaces, cgroups, union filesystems, OS-level vs hardware virtualization 9 Docker and the Container Ecosystem – images and layers, the OCI spec, registries, writing a sound Dockerfile	Namespaces and cgroups by hand; a first Dockerfile	Assignment 1 set
8	10 Container Orchestration with Kubernetes – pods, deployments and services, scheduling, self-healing, scaling	Midterm, then a Deployment and a Service on kind	Midterm (Ch 1–9)
9	11 Lightweight and Secure Virtualization – microVMs (Firecracker), gVisor, Kata Containers, unikernels	A Firecracker microVM, boot time measured	Project set Assignment 1 due
10	12 Storage Virtualization – SAN/NAS abstraction, virtual disks, thin provisioning, software-defined storage, Ceph/vSAN 13 Network Virtualization and SDN – virtual switches, VLANs and VXLAN overlays, SDN controllers, NFV	A thin-provisioned pool; a small Ceph cluster	Project proposal due
11	14 Cloud Service and Deployment Models – IaaS/PaaS/SaaS/FaaS, public/private/hybrid/multi-cloud, VM instances on AWS/Azure/GCP	One workload, three clouds: AWS, Azure, GCP instances	Quiz 2 (Ch 10–14)
12	15 Desktop and Application Virtualization – VDI, Citrix/Horizon, application streaming, remote browser isolation 16 Virtualization Management Platforms – vCenter, Hyper-V Manager/SCVMM, oVirt, Proxmox, OpenStack, Terraform	A VDI session; a streamed application	Assignment 2 set
13	17 High Availability, Live Migration and DR – vMotion/Live Migration, clustering, snapshots, checkpointing, backup strategy	Live Migration with zero dropped pings	Project experiments frozen
14	18 Performance Monitoring and Capacity Planning – hypervisor metrics, the noisy-neighbour effect, benchmarking virtualized workloads 19 Security in Virtualized Environments – VM escape, hypervisor attack surface, Spectre/Meltdown in multi-tenant clouds, container hardening	fiio/stress-ng: native vs virtualized; a container break-out drill	Quiz 3 (Ch 15–19) Assignment 2 due
15	20 GPU and Accelerator Virtualization – GPU passthrough, vGPU and MIG partitioning, virtualizing AI/ML workloads 21 Confidential Computing and Emerging Hardware – Intel TDX, AMD SEV, trusted execution environments, composable infrastructure	A vGPU/MIG partition for a training job	—
16	22 Case Studies in Enterprise and Cloud Virtualization – data-centre consolidation, cloud migration, edge virtualization 23 Semester Project Presentations – deployed-service demos, peer review, course review	Peer walkthroughs of each team's deployed service	Project report and presentation

ASSESSMENT WEIGHTS

20%	30%	50%
Sessional: quizzes, assignments, project	Mid-term, Week 8	Final-term, all chapters

I Foundations of Virtualization**IV** Storage, Network and Cloud Virtualization**II** Hypervisors and Compute Virtualization**V** Management, Performance and Security**III** Containers and Lightweight Virtualization**VI** Emerging Hardware and Practice

Modernised from HEC 2017 (*Curriculum of BS & MS CS, SE & IT*, pp. 71–130): the original hypervisor trio — VMware, Xen and Hyper-V — opens the course in Weeks 3–6; Weeks 7–16 carry the same ground into containers, Kubernetes, cloud models and confidential computing, anticipating HEC 2025's *Cloud Computing and Microservices Architecture* and *Docker Containers*, the two courses that replaced it. The 2017 set texts (Stanney, Burdea) describe virtual reality, not system virtualization, and are not used here. Chapter numbers refer to the course handout, *Virtual Systems and Services* — *From Hypervisors to the Cloud-Native Stack* (239 pp., 23 chapters, 7 appendices); Appendix G of that handout is the source of record for this page. Weights follow the department's standard split: Sessional 20%, Mid-term 30%, Final-term 50%.